

Virtual Welfare

Imagine a single buyer will arrive with their private value v , drawn from a known continuous distribution F . We want to design DSIC mechanisms.

What mechanism should you use to maximize *welfare* ($\sum_i v_i x_i$) Always give the bidder the item, always give it away for free!

What should you do to maximize (expected) revenue? Post a price that maximizes $\text{REV} = \max_r r \cdot [1 - F(r)]$.

Definition 1. In a deterministic mechanism, given other bids $\mathbf{b}_{-i}$, bidder i 's *critical bid* is the minimum bid $b_i^* = \min\{b_i : x_i(b_i, \mathbf{b}_{-i}) = 1\}$ such that bidder i is allocated to.

Then with $\mathbf{b}_{-i}$ fixed, for all winning $v_i \geq b_i^*$, i 's payment $p_i(v_i, \mathbf{b}_{-i}) = b_i^*$ is their critical bid.

What is winner i 's critical bid in a single-item auction? The second-highest bid!

What about in the k identical item setting? The $k + 1^{\text{st}}$ bid!

Revenue Maximization and Myersonian Virtual Welfare

Recall:

- The revelation principle says that it's without loss to focus only on truthful mechanisms.
- Payment is determined by the allocation:

$$p_i(b_i, \mathbf{b}_{-i}) = b_i \cdot x_i(b_i, \mathbf{b}_{-i}) - \int_0^{b_i} x_i(z, \mathbf{b}_{-i}) dz$$

For this lecture, values are independent and nonnegative. Each F_i has a density f_i , positive on the interior of its support, and finite expected value. Payments are normalized so that a bidder of value zero pays zero.

Fixed, choice, and randomness.

- **Fixed:** the distributions and the other bids $\mathbf{v}_{-i}$; for the derivation, also fix an allocation rule x .
- **Choice:** the designer chooses the allocation rule x ; its payments are then determined by the payment identity.
- **Randomness:** first $v_i \sim F_i$; afterward, average over $\mathbf{v}_{-i}$ as well.

Probability for the revenue calculation

If V is nonnegative with density f and CDF F , then

$$\mathbb{E}[g(V)] = \int_0^\infty g(t)f(t) dt, \quad \Pr[V \geq t] = 1 - F(t) = \int_t^\infty f(s) ds.$$

The first formula averages any integrable function $g(V)$, not just V itself. For a continuous distribution, including the endpoint does not change a probability.

Changing the order of integration. The region $0 \leq s \leq t$ can be described in two ways: fix t and let s run from 0 to t , or fix s and let t run from s to ∞ . For nonnegative h ,

$$\int_0^\infty \int_0^t h(s, t) ds dt = \int_0^\infty \int_s^\infty h(s, t) dt ds.$$

Both sides add up the same contributions. The inner integral is evaluated first.

Averaging in stages. For independent V, W , first fix $W = w$ and average $H(V, w)$ over V , then average the resulting function over W :

$$\mathbb{E}_W[\mathbb{E}_V[H(V, W)]] = \mathbb{E}_{V, W}[H(V, W)].$$

Independence keeps the distribution of V unchanged when W is fixed. Here $V = v_i$ and $W = \mathbf{v}_{-i}$. Also, $\mathbb{E}[\sum_i Z_i] = \sum_i \mathbb{E}[Z_i]$ by linearity, without needing independence.

We want to maximize $\mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[\sum_i p_i(\mathbf{v})]$.

$$\begin{aligned} \mathbb{E}_{v_i \sim F_i}[p_i(v_i, \mathbf{v}_{-i})] &= \int_0^\infty f_i(v_i) p_i(v_i, \mathbf{v}_{-i}) dv_i \\ &= \int_0^\infty f_i(v_i) \left[v_i \cdot x_i(v_i, \mathbf{v}_{-i}) - \int_0^{v_i} x_i(z, \mathbf{v}_{-i}) dz \right] dv_i \\ &= \int_0^\infty \left[f_i(v_i) v_i x_i(v_i, \mathbf{v}_{-i}) - x_i(v_i, \mathbf{v}_{-i}) \left[\int_{v_i}^\infty f_i(z) dz \right] \right] dv_i \quad (*) \\ &= \int_0^\infty \left[f_i(v_i) v_i x_i(v_i, \mathbf{v}_{-i}) - x_i(v_i, \mathbf{v}_{-i}) [1 - F_i(v_i)] \right] dv_i \\ &= \int_0^\infty f_i(v_i) x_i(v_i, \mathbf{v}_{-i}) \left[v_i - \frac{[1 - F_i(v_i)]}{f_i(v_i)} \right] dv_i \\ &= \mathbb{E}_{v_i \sim F_i}[\varphi_i(v_i) x_i(v_i, \mathbf{v}_{-i})] \end{aligned}$$

where

$$\varphi_i(v_i) = v_i - \frac{[1 - F_i(v_i)]}{f_i(v_i)}$$

is the Myersonian virtual value and $(*)$ follows by switching the order of integration. Then

$$\text{REVENUE} = \mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[\sum_i p_i(\mathbf{v})] = \sum_i \mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[p_i(\mathbf{v})] = \sum_i \mathbb{E}_{\mathbf{v} \sim \mathbf{F}}[\varphi_i(v_i) x_i(v_i, \mathbf{v}_{-i})]$$

Take $\mathbb{E}_{\mathbf{v}_{-i} \sim \mathbf{F}_{-i}}$ of both sides of the previous equation. By averaging in stages, the nested expectations become expectations over the full profile $\mathbf{v} \sim \mathbf{F}$.

$$= \mathbb{E}_{\mathbf{v} \sim \mathbf{F}} \left[\sum_i \varphi_i(v_i) x_i(\mathbf{v}) \right] = \text{VIRTUAL WELFARE}$$

Given this conclusion, how should we design our allocation rule x to maximize expected virtual welfare (expected revenue)? Give the item to the bidder with the highest *nonnegative virtual* value, if any!

When would this cause a problem with incentive-compatibility? When the corresponding x isn't monotone!

Definition 2. A distribution F is regular if the corresponding virtual valuation function $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ is strictly increasing.

Suppose we are in the single-item setting and all of the distributions are regular. What do the payments look like in the virtual-welfare-maximizing allocation?

For a fixed $\mathbf{b}_{-i}$, if i is the winner, then i 's payment is i 's critical bid, which is $\varphi_i^{-1}(\tau_i)$ where $\tau_i = \max\{0, \max_{j \neq i} \varphi_j(b_j)\}$ is the highest competing virtual value or zero, whichever is larger. Exercise: what about for k identical items?

Claim 1. A virtual welfare maximizing allocation x is monotone if the virtual value functions are regular, with a fixed priority order for breaking ties.

Exercise: Argue this.

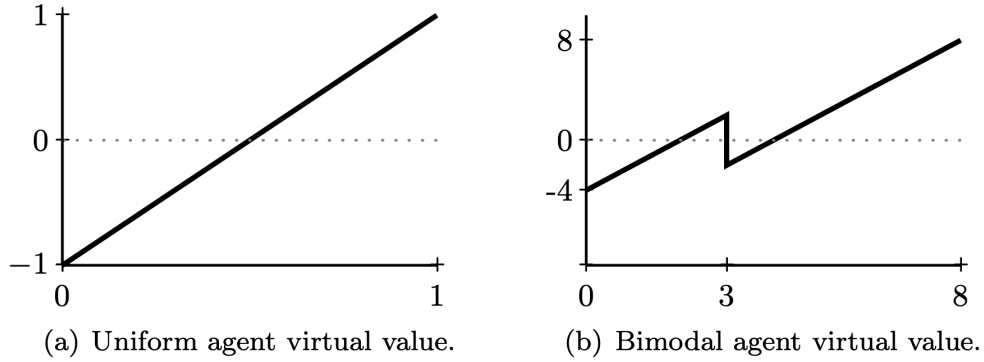


Figure 1: Virtual value functions $\varphi(v) = v - \frac{1-F(v)}{f(v)}$ for the uniform and bimodal agent examples.

It will be helpful to keep the following two examples in mind:

- a. a uniform agent with $v \sim U[0, 1]$. Then $F(x) = x$ and $f(x) = 1$.
- b. a bimodal agent with

$$v \sim \begin{cases} U[0, 3] & w.p. \frac{3}{4} \\ U(3, 8) & w.p. \frac{1}{4} \end{cases} \quad \text{and} \quad f(v) = \begin{cases} \frac{1}{4} & v \in [0, 3] \\ \frac{1}{20} & v \in (3, 8] \end{cases}$$

Do the following:

- Calculate the virtual values for both examples.

a. $\varphi(v) = 2v - 1$

b. $1 - F(v) = \begin{cases} \frac{1}{4} + \left(\frac{3-v}{3}\right) \cdot \frac{3}{4} & v \in [0, 3] \\ \left(\frac{8-v}{5}\right) \cdot \frac{1}{4} & v \in (3, 8] \end{cases}$ so $\varphi(v) = \begin{cases} 2v - 4 & v \in [0, 3] \\ 2v - 8 & v \in (3, 8] \end{cases}$

- Are they regular? Are there any issues using the allocation that maximizes expected virtual welfare?

a. Yep!

b. Nope. As we can see in Figure ??, $\varphi(3.5) = -1 < \varphi(2.5) = 1$. This implies a single bidder gets allocated with $v = 2.5$ but then stops getting allocated as they increase their value to 3.5.

- What does that allocation actually look like?

a. For a single bidder, allocate above $v = 0.5$ at a price (critical bid) of $\varphi^{-1}(0) = 0.5$.

b. The virtual welfare maximizing allocation isn't DSIC! Turns out you can do something to make φ monotone and *then* use the VW-maximizing allocation. We'll do this later in class.

Quantile Space

Position in the distribution. Assume F is continuous and strictly increasing on its support. An agent's *quantile* is the fraction of the distribution with higher values: $q = 1 - F(v)$. We let $v(q)$ denote the corresponding value.

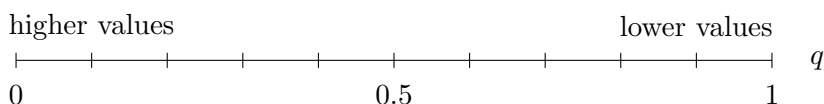
Consider a distribution that is $U[\$0, \$10]$. A value of \$9 is in the top 10%, so its quantile is 0.1 and $v(0.1) = \$9$. A value of \$2 has quantile $1 - 2/10 = 0.8$. Smaller quantiles correspond to higher values.

Quantile distribution. Let $V \sim F$ and let $Q = 1 - F(V)$ be its quantile. The probability that a random draw is in the top 20% is 0.2; the probability that it is between the top 20% and the top 50% is $0.5 - 0.2 = 0.3$.

More generally, $Q \leq a$ means that the draw is in the top a fraction of the distribution. Therefore

$$\Pr[Q \leq a] = a \quad \text{for } 0 \leq a \leq 1, \quad Q \sim U[0, 1].$$

Each interval below contains 10% of the probability. The corresponding intervals in value space need not have equal lengths.



Averaging. Suppose $g(v)$ is a quantity we want to average, such as a payment. At quantile q , that same quantity is $g(v(q))$. Since quantiles have density 1 on $[0, 1]$,

$$\mathbb{E}[g(V)] = \int_0^\infty g(v)f(v) dv = \int_0^1 g(v(q)) dq.$$

Here we assume the expectation exists. Equal-width intervals in quantile space have equal probability, so the density factor is already accounted for. The distribution F still determines the values $v(q)$.

Revenue. A posted price of $v(q)$ sells with probability q . Its expected revenue is

$$R(q) = q v(q).$$

For $U[\$0, \$10]$, posting \$9 sells with probability 0.1 and gives expected revenue \$0.90.

Now draw the posted price from F , independently of the buyer's value. Its quantile is uniform too. Conditional on a price with quantile q , the expected revenue is $R(q)$. Averaging over the price gives

$$\text{Expected revenue} = \int_0^1 R(q) dq.$$

This is the area under the revenue curve in quantile space.

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